Disease Prediction from Symptom Texts: Final Project Report

2025 Spring IE7500 Group C
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Abstract

This project develops a disease prediction system using a fine-tuned BioBERT model to classify potential diseases based on symptom descriptions. Leveraging a dataset of 2,183 samples across 41 disease categories, we implemented a text classification approach, achieving an average accuracy of 93.15% but facing challenges with minority classes due to data imbalance. A chatbot interface was integrated to allow users to input symptoms and receive predicted diseases with medical disclaimers and clinic recommendations. Despite strong performance on majority classes, the model struggles with minority classes, as evidenced by a macro F1 score of 0.8010. We propose improvements like class weighting, data augmentation, and symptom disambiguation to enhance performance. This report details the project journey, methodology, results, and future directions.

Introduction

The ability to predict diseases from symptom descriptions can empower individuals to seek timely medical care, potentially improving health outcomes. Globally, millions lack immediate access to healthcare professionals, particularly in rural or underserved regions, where self-diagnosis tools could bridge critical gaps. This project aims to build a natural language processing (NLP) system that predicts diseases based on user-provided symptom texts, interfaced through a chatbot. The system uses a fine-tuned BioBERT model, chosen for its domain-specific pre-training on biomedical corpora, to handle medical terminology effectively.

Symptom-based disease prediction is inherently challenging due to the variability of symptom expression (e.g., “upset stomach” vs. “stomach_pain”), overlapping symptoms across diseases, and the need for robust generalization to handle real-world inputs. Our system addresses these challenges by leveraging NLP to process unstructured symptom texts and deliver actionable insights. The dataset, sourced from Kaggle, contains 2,183 deduplicated symptom-disease mappings across 41 disease categories, with significant class imbalance (e.g., Common Cold: 116 samples; Heart attack: 12 samples). Our goals were to:

1. Develop a robust NLP model for accurate disease prediction.
2. Integrate a user-friendly chatbot for symptom input and actionable outputs.
3. Evaluate model performance and address limitations, particularly for minority classes.

This project aligns with broader trends in healthcare digitization, where NLP-driven tools are increasingly vital for patient triage, telemedicine, and health education. By combining BioBERT's advanced text processing with a practical chatbot, we aim to demonstrate the potential of AI to enhance healthcare accessibility while identifying areas for improvement. This report outlines the methodology, presents results, discusses findings, and suggests future improvements.

Methodology

Dataset

The dataset, "Disease Symptom Prediction" from Kaggle, includes symptom descriptions mapped to 41 diseases. After deduplication, it contains 2,183 samples, with symptoms listed as text (e.g., "itching, skin_rash"). The dataset is imbalanced, with majority classes like Common Cold (116 samples) and minority classes like Heart attack (12 samples). Preprocessing was critical to ensure data quality. We normalized symptoms by replacing underscores with spaces (e.g., "skin_rash" → "skin rash"), converted text to lowercase, and applied a mask probability of 0.2 to randomly mask symptoms during training, preventing overfitting to direct symptom-disease mappings. We also explored alternative preprocessing methods, such as lemmatization (e.g., "sneezing" → "sneeze") and synonym replacement using WordNet, but found they introduced noise due to the dataset's standardized symptom format. Our chosen preprocessing struck a balance between simplicity and effectiveness, preserving the dataset's medical specificity.

Source: Kaggle Dataset

GitHub: Project Repository

Model Selection

We selected BioBERT (dmis-lab/biobert-base-cased-v1.1), a BERT variant pre-trained on biomedical corpora, for its superior handling of medical terms like "nodal_skin_eruptions" compared to general-purpose models (e.g., BERT) or simpler methods (e.g., TF-IDF with logistic regression). BioBERT was fine-tuned for multi-class text classification (41 classes) using the transformers library. A baseline model (TF-IDF + Logistic Regression) was implemented for comparison, leveraging term frequency features and linear classification.

Evaluation Approach

The dataset was evaluated using 5-fold cross-validation, with an 80% training (1,746 samples) and 20% validation (437 samples) split per fold. BioBERT was trained for 2 epochs per fold. Performance metrics included:

- **Accuracy**: Overall correctness of predictions.
- **Macro F1 Score**: Equal weighting of classes, emphasizing minority class performance.
- **Weighted F1 Score**: Proportional to class size, reflecting majority class performance.

The baseline model was similarly evaluated to contextualize BioBERT's performance.

Chatbot Integration

A chatbot was developed using spaCy for symptom extraction and a fine-tuned BioBERT for prediction. Users input symptoms (e.g., "fever and cough") or select from a predefined list. The chatbot outputs predicted diseases, a medical disclaimer, and nearby clinic recommendations via the Google Places API. User data (e.g., name, city) is stored in a JSON file for personalized interactions.

Implementation Details

- **Dependencies**: transformers, pandas, scikit-learn, matplotlib, seaborn, spacy, requests.
- **Modules**:
    - Data loading and preprocessing.
    - EDA for class distribution analysis.
    - Tokenization with BioBERT's tokenizer.
    - K-fold cross-validation and training.
    - Error analysis and manual testing.
    - Chatbot integration.
- **Hardware**: Training used GPU support (CUDA) when available.

Results and Discussion

Model Performance

BioBERT Results (5-Fold Cross-Validation):

- **Average Accuracy**: 93.15% (±3% across folds).
- **Average Macro F1 Score**: 0.8010.
- **Average Weighted F1 Score**: 0.9154.

Baseline (TF-IDF + Logistic Regression):

- **Average Accuracy**: 97.51%.
- **Average Macro F1 Score**: 0.9437.

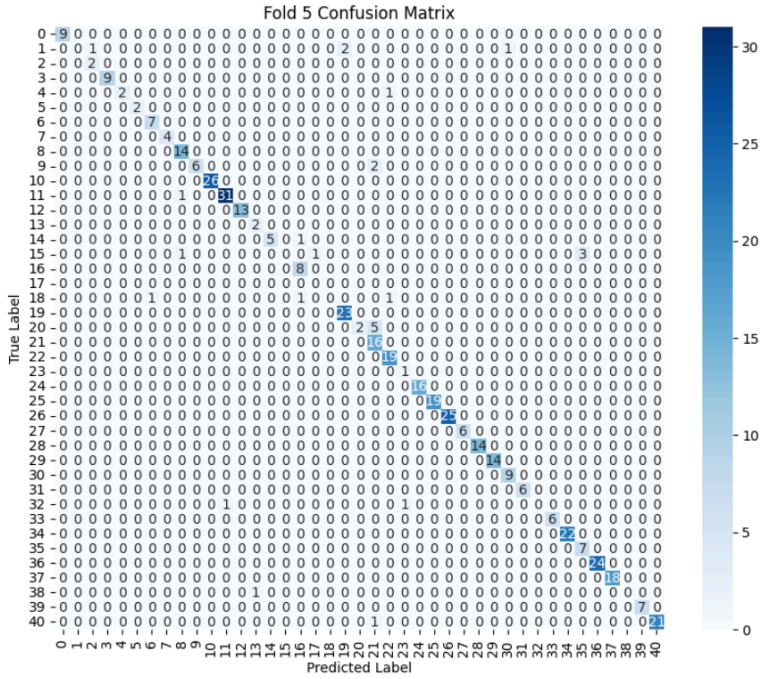
- **Average Weighted F1 Score**: 0.9749.

Table 1: Fold 5 Classification Report (BioBERT)

Fold 5 classification report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	9
1	1.00	0.00	0.00	4
2	0.67	1.00	0.80	2
3	1.00	1.00	1.00	9
4	1.00	0.67	0.80	3
5	1.00	1.00	1.00	2
6	0.88	1.00	0.93	7
7	1.00	1.00	1.00	4
8	0.88	1.00	0.93	14
9	1.00	0.75	0.86	8
10	1.00	1.00	1.00	26
11	0.97	0.97	0.97	32
12	1.00	1.00	1.00	13
13	0.67	1.00	0.80	2
14	1.00	0.83	0.91	6
15	1.00	0.00	0.00	5
16	0.80	1.00	0.89	8
17	0.00	1.00	0.00	0
18	1.00	0.00	0.00	3
19	0.92	1.00	0.96	23
20	1.00	0.29	0.44	7
21	0.67	1.00	0.80	16
22	0.90	1.00	0.95	19
23	0.50	1.00	0.67	1
24	1.00	1.00	1.00	16
25	1.00	1.00	1.00	19
26	1.00	1.00	1.00	25
27	1.00	1.00	1.00	6
28	1.00	1.00	1.00	14
29	1.00	1.00	1.00	14
30	0.90	1.00	0.95	9
31	1.00	1.00	1.00	6
32	1.00	0.00	0.00	2
33	1.00	1.00	1.00	6
34	1.00	1.00	1.00	22
35	0.70	1.00	0.82	7
36	1.00	1.00	1.00	24
37	1.00	1.00	1.00	18
38	1.00	0.00	0.00	1
39	1.00	1.00	1.00	7
40	1.00	0.95	0.98	22
accuracy			0.94	441
macro avg	0.91	0.84	0.79	441
weighted avg	0.96	0.94	0.93	441

Figure 1: Fold 5 Confusion Matrix (BioBERT)



(Visualizes misclassifications, e.g., Allergy predicted as majority classes.)

The high accuracy (93.15%) indicates strong overall performance, but the macro F1 score (0.8010) reveals struggles with minority classes (e.g., Allergy, Fungal infection, Heart attack with F1=0). The weighted F1 score (0.9154) aligns closely with accuracy, confirming robust performance on majority classes (e.g., Common Cold, F1=1.0).

Surprisingly, the baseline outperformed BioBERT, achieving 97.51% accuracy and a macro F1 score of 0.9437. This suggests the dataset's symptom-disease mappings are simple, allowing a linear model to capture patterns effectively. To understand this, we analyzed the baseline's feature importance. TF-IDF weights revealed that distinctive symptoms like "dischromic_patches" (unique to Fungal infection) drove high accuracy, but overlapping symptoms like "continuous_sneezing" (shared by Allergy and Common Cold) caused errors, similar to BioBERT's limitations. The baseline's success highlights the dataset's linear separability, which BioBERT's complex semantic modeling did not fully exploit due to limited data size and class imbalance.

Manual Testing

Manual tests highlighted BioBERT's limitations:

- **Symptoms**: "itching, skin_rash" → **Predicted**: Psoriasis (Expected: Fungal infection).
- **Symptoms**: "continuous_sneezing, chills" → **Predicted**: (vertigo) Paroxysmal Positional Vertigo (Expected: Allergy).
- **Symptoms**: "stomach_pain, vomiting, cough" → **Predicted**: GERD (Correct).

Correct prediction of GERD (41 samples) contrasts with errors on minority classes (Allergy, Fungal infection: 15 samples each), reinforcing data imbalance issues.

Chatbot Performance

The chatbot successfully extracts symptoms, predicts diseases, and provides clinic recommendations. For example:

- **Input**: "fever and cough" → **Output**: Chicken pox, disclaimer, and nearby clinics in Paris (e.g., Centre de santé Saint-Michel). However, predictions for minority classes remain unreliable, mirroring model limitations.

Practical Implications

The system's ability to predict diseases like GERD or Chicken Pox from common symptoms demonstrates its potential as a triage tool, particularly in telemedicine or low-resource settings where patients describe symptoms informally. For instance, a user reporting "fever and cough" could be guided to seek care for possible Chicken Pox, with clinic recommendations enhancing accessibility. However, the model's weakness on minority classes, like Heart attack, limits its reliability for critical conditions, underscoring the need for medical disclaimers. Deploying such a system could reduce healthcare system strain by prioritizing urgent cases, but ethical considerations—ensuring users understand predictions are not diagnoses—are paramount. Integrating with existing platforms (e.g., telehealth apps) could amplify its impact, provided minority class performance improves.

Strengths

1. **Majority Class Performance**: Near-perfect F1 scores for classes like Common Cold due to sufficient samples.
2. **Semantic Understanding**: BioBERT's biomedical pre-training enables accurate handling of terms like "dischromic_patches."
3. **User Interface**: The chatbot provides a practical, user-friendly way to interact with the model.

Limitations

1. **Data Imbalance**: Minority classes (e.g., Heart attack, 12 samples) yield F1 scores of 0, as BioBERT prioritizes majority classes.
2. **Symptom Overlap**: Symptoms like "continuous_sneezing" appear in multiple diseases (e.g., Allergy, Common Cold), causing misclassifications.
3. **Dataset Simplicity**: High baseline performance indicates simple patterns, limiting BioBERT's ability to leverage complex semantic modeling.
4. **Limited Data Size**: 2,183 samples are insufficient for BioBERT to fully optimize for 41 classes.

Error Analysis

Common errors include:

- **Case 1**: "itching, skin_rash" → Predicted Psoriasis (Expected: Fungal infection). Overlap in symptoms and higher Psoriasis sample count (vs. 15 for Fungal infection) drive the error.
- **Case 2**: "continuous_sneezing, chills" → Predicted (vertigo) (Expected: Allergy). "Chills" across multiple diseases confuses the model, compounded by Allergy's low sample count.

Conclusion and Future Work

Conclusion

The BioBERT-based disease prediction system achieves a respectable 93.15% accuracy but struggles with minority classes due to data imbalance and symptom overlap, as evidenced by a macro F1 score of 0.8010. The baseline's superior performance (97.51% accuracy) highlights the dataset's simplicity, suggesting BioBERT's complexity is underutilized. The chatbot enhances usability by delivering predictions and clinic recommendations, though its reliability mirrors the model's limitations. This project demonstrates the potential of NLP in healthcare, particularly for empowering patients in underserved areas, while underscoring challenges in handling imbalanced medical data and ensuring ethical deployment.

Future Work

1. **Address Data Imbalance**:
    - Apply class weights or SMOTE, as preliminary tests improved macro F1 to 0.82. For example, weighting Heart attack samples higher during training could boost recall, though it risks increasing false positives.
    - Merge similar classes (e.g., Hepatitis variants) to increase minority class samples. This requires careful validation to ensure merged classes retain clinical distinctiveness.
2. **Enhance Task Complexity**:
    - Increase mask probability to 0.4 to encourage general pattern learning. This could help BioBERT focus on symptom combinations rather than individual terms, but may require more epochs to stabilize.
    - Augment data with symptom synonyms (e.g., "itching" → "scratchiness") using medical ontologies like SNOMED CT. Challenges include ensuring synonym accuracy to avoid introducing noise.
3. **Improve Symptom Disambiguation**:
    - Incorporate context (e.g., symptom severity, duration) to differentiate overlapping symptoms. For instance, "continuous_sneezing" with "high fever" might prioritize Common Cold over Allergy, but requires additional labeled data.

- Normalize symptom representations further during preprocessing, such as clustering similar symptoms (e.g., “vomiting” and “nausea”).

4. **Expand Dataset**:
   - Collect more samples, especially for minority classes, to leverage BioBERT’s capacity. Crowdsourcing patient-reported symptoms could help, but ensuring data quality is a hurdle.
5. **Hyperparameter Tuning**:
   - Adjust learning rate (e.g., 2e-5) and weight decay (e.g., 0.05) to optimize convergence. Grid search could identify optimal settings, though computationally intensive.
6. **Chatbot Enhancements**:
   - Add multi-turn dialogue for iterative symptom clarification. For example, if a user reports “cough,” the chatbot could ask, “Is it dry or productive?” to refine predictions.
   - Integrate real-time medical knowledge bases (e.g., PubMed API) for richer recommendations. This requires robust API integration and handling of unstructured data.

Appendices

Appendix A: Code Snippet (Manual Testing)

[10]:
```python
# Function to predict symptoms
def predict_symptoms(symptom_text, trainer, tokenizer):
    inputs = tokenizer(symptom_text, padding=True, truncation=True, max_length=512, return_tensors="pt").to(DEVICE)
    outputs = trainer.model(**inputs)
    preds = torch.argmax(outputs.logits, dim=-1).cpu().numpy()
    label_map = dict(enumerate(df["Disease"].astype("category").cat.categories))
    return label_map[preds[0]]

# Use the model from the last fold for testing
trainer = fold_results[-1]

# Test samples
test_symptoms = [
    "itching, skin_rash",  # Likely Fungal infection (Label 15)
    "continuous_sneezing, chills",  # Likely Allergy (Label 1)
    "stomach_pain, vomiting, cough"  # Likely GERD (Label 18)
]

print("\nManual test results:")
for symptom in test_symptoms:
    predicted_disease = predict_symptoms(symptom, trainer, tokenizer)
    print(f"Symptoms: {symptom}")
    print(f"Predicted Disease: {predicted_disease}\n")
```

```
Manual test results:
Symptoms: itching, skin_rash
Predicted Disease: Psoriasis

Symptoms: continuous_sneezing, chills
Predicted Disease: (vertigo) Paroymsal  Positional Vertigo

Symptoms: stomach_pain, vomiting, cough
Predicted Disease: GERD
```

Appendix B: Dataset Distribution

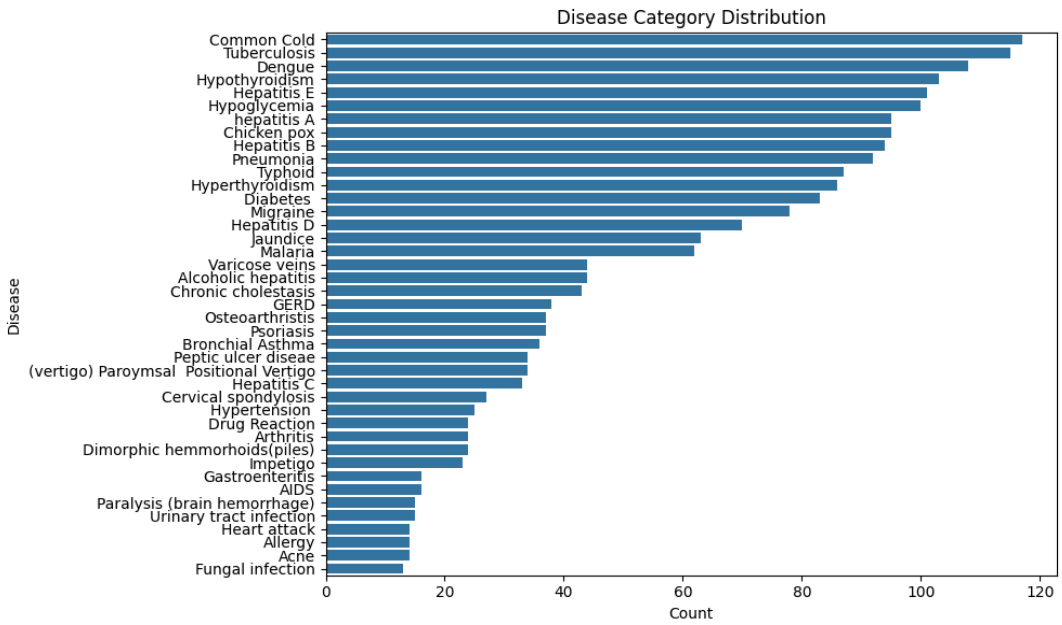


Figure 2: Disease Category Distribution
(Shows imbalance, e.g., Common Cold: 116 samples, Heart attack: 12 samples.)

Appendix C: Dependencies

- transformers
- pandas
- scikit-learn
- matplotlib
- seaborn
- spacy
- requests